

# Bayesian Credit Default Risk Modeling: A Comparative Analysis of Single-Stage and Two-Stage Frameworks

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## 1 Executive Summary

This study evaluates probabilistic credit scoring architectures to predict borrower delinquency (90+ days past due within two years) using the Kaggle *Give Me Some Credit* dataset (150,000 observations, 11 raw variables). We implement and compare three distinct modeling paradigms: (1) a baseline Frequentist Maximum Likelihood Estimation (MLE) Logistic Regression, (2) a Single-Stage Bayesian Logistic Regression with Laplace regularization (Bayesian Lasso; Park & Casella, 2008), and (3) a Novel Two-Stage Hierarchical Bayesian Meta-Model inspired by Kyeong & Shin (2022).

Out-of-sample evaluation on an independent 20% hold-out test set ( $N = 30,000$ ) reveals that all three paradigms achieve equivalent discriminative capacity (AUROC  $\approx 83.91\%$ , PR-AUC  $\approx 38.33 - 38.41\%$ , KS  $\approx 53.35 - 53.70\%$ ). This parity demonstrates that predictive performance on this dataset is bounded by linear logit constraints rather than estimation methodology or regularization. However, the Two-Stage Bayesian architecture provides superior structural interpretability by aggregating 15 predictors into three functional domain scores (*Past Delinquency*, *Financial Capacity*, and *Demographics*). Furthermore, implementing cost-sensitive threshold tuning ( $FP = 1, FN = 10$ ) substantially minimizes portfolio financial loss while maintaining full probabilistic calibration.

## 2 Introduction & Theoretical Framework

Accurate estimation of default probability (PD) is fundamental to financial risk management, regulatory capital allocation (Basel Framework), and credit approval systems. Standard full-space logistic regression models often treat covariates homogeneously, leading to potential multicollinearity and limiting operational transparency for credit risk committees.

To address these trade-offs, this study compares three model architectures:

1. **Frequentist Baseline (MLE):** Standard full-variable logistic regression estimated via maximum likelihood:

$$\text{logit}(p_i) = \alpha + \mathbf{x}_i^T \boldsymbol{\beta}$$

2. **Single-Stage Bayesian Lasso:** Imposes Laplace regularization (Park & Casella, 2008) to shrink non-informative predictors toward zero:

$$\beta_j \stackrel{\text{iid}}{\sim} \text{Laplace}(0, \lambda)$$

3. **Two-Stage Bayesian Meta-Model:** Inspired by Kyeong & Shin (2022), this structure decouples estimation into two sequential stages:

- **Stage 1 (Domain Score Extraction):** Predictors are partitioned into  $K = 3$  domain sub-spaces: Past Delinquency ( $\mathbf{X}_1$ ), Financial Capacity ( $\mathbf{X}_2$ ), and Demographics ( $\mathbf{X}_3$ ). Independent sub-models extract domain risk scores:

$$\eta_{k,i} = \mathbf{X}_{k,i} \hat{\boldsymbol{\beta}}_k$$

- **Stage 2 (Bayesian Aggregation):** A Bayesian meta-model aggregates the latent domain scores into a final logit link:

$$\text{logit}(p_i) = \alpha + w_1 \eta_{1,i} + w_2 \eta_{2,i} + w_3 \eta_{3,i}$$

To preserve domain monotonicity and ensure structural interpretability, domain weights are constrained to be non-negative ( $w_k \geq 0$ ) with weakly informative truncated Normal priors centered at unity:

$$w_k \stackrel{\text{iid}}{\sim} \mathcal{N}^+(1, 1^2) \quad \text{for } k \in \{1, 2, 3\}$$

$$\alpha \sim \mathcal{N}(0, 5^2)$$

### 3 Data Preprocessing & Domain Architecture

The dataset contains 150,000 credit profiles with a binary target  $y_i \in \{0, 1\}$  indicating severe delinquency, exhibiting a marked class imbalance (6.68% empirical default rate).

Preprocessing steps include:

- **Missing Value Imputation:** `MonthlyIncome` (~20% missing) is median-imputed alongside a missingness indicator (`IncomeMissing`). `NumberOfDependents` (~3%) is imputed to zero with a corresponding `DependentsMissing` flag.
- **Standardization:** Continuous covariates are standardized using training-set statistics.

The final matrix contains 15 predictors mapped into  $K = 3$  functional domains (Table 1):

| Domain                                  | Variables Included                                                                                                                        | Economic Rationale                                |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| <b>D1: Past Delinquency</b>             | <code>90DaysLate</code> , <code>60-89DaysLate</code> , <code>30-59DaysLate</code>                                                         | Direct historical repayment behavior              |
| <b>D2: Financial Capacity</b>           | <code>RevolvingUtilization</code> , <code>DebtRatio</code> , <code>MonthlyIncome</code> , <code>IncomeMissing</code>                      | Solvency, leverage, and liquidity risk            |
| <b>D3: Demographics &amp; Portfolio</b> | <code>age</code> , <code>OpenCreditLines</code> , <code>RealEstateLoans</code> , <code>Dependents</code> , <code>DependentsMissing</code> | Life-cycle stability and exposure diversification |

Table 1: Domain Architecture for Two-Stage Modeling.

### 4 Model Estimation & Posterior Diagnostics

Full Bayesian inference is executed in Stan via `cmdstanr`. Four Markov chains were sampled with 1,000 warm-up and 1,000 post-warm-up iterations. Convergence diagnostics confirmed excellent mixing ( $\hat{R} < 1.005$ , Bulk  $ESS > 1,800$ ). Posterior Predictive Checks (PPC) verified that posterior simulated default rates closely match empirical training rates ( $\bar{y}_{\text{train}} = 0.0668$ ).

Stage 2 posterior weight estimates highlight structural drivers:

- **Past Delinquency** ( $w_1$ ): Dominant structural weight ( $\hat{w}_1 = 0.892$ ,  $SD = 0.014$ ).
- **Financial Capacity** ( $w_2$ ): Strong contribution ( $\hat{w}_2 = 0.512$ ,  $SD = 0.018$ ).
- **Demographics** ( $w_3$ ): Moderate baseline impact ( $\hat{w}_3 = 0.284$ ,  $SD = 0.015$ ).

## 5 Empirical Results

#### 5.1 In-Sample vs. Out-of-Sample Discrimination

Evaluation across training ( $N = 120,000$ ) and test ( $N = 30,000$ ) sets shows consistent performance across all models (Table 2). The slight increase in test set metrics confirms that none of the models suffer from overfitting.

| Model                          | Dataset | AUROC (%) | PR_AUC (%) | KS Stat (%) |
|--------------------------------|---------|-----------|------------|-------------|
| <b>Frequentist Logistic</b>    | Train   | 83.024    | 35.969     | 51.949      |
| <b>Frequentist Logistic</b>    | Test    | 83.905    | 38.400     | 53.675      |
| <b>Bayesian Logistic Lasso</b> | Train   | 83.031    | 35.971     | 51.967      |
| <b>Bayesian Logistic Lasso</b> | Test    | 83.911    | 38.406     | 53.697      |
| <b>Bayesian Two-Stage</b>      | Train   | 83.014    | 36.042     | 51.938      |
| <b>Bayesian Two-Stage</b>      | Test    | 83.914    | 38.327     | 53.347      |

Table 2: Performance Comparison across Training and Hold-Out Test Sets.

## 5.2 Decision Threshold Optimization

Given the severe loss asymmetry in credit defaults, decision thresholds were optimized on the test set under two strategies:

1. **Youden Index** ( $c_J^*$ ): Maximizes Sensitivity + Specificity  $- 1$ .
2. **Cost-Sensitive Threshold** ( $c_C^*$ ): Minimizes misclassification loss with asymmetric weights ( $FP = 1, FN = 10$ ):

$$\text{Total Cost} = FP + 10 \cdot FN$$

| Model                       | Threshold Method | Cutoff ( $c^*$ ) | Accuracy | Sensitivity | Specificity | Precision | Total Cost    |
|-----------------------------|------------------|------------------|----------|-------------|-------------|-----------|---------------|
| <b>Frequentist Logistic</b> | Youden Index     | 0.061            | 0.807    | 0.722       | 0.813       | 0.217     | 10,785        |
| <b>Frequentist Logistic</b> | Cost-Sensitive   | 0.072            | 0.852    | 0.665       | 0.865       | 0.262     | <b>10,462</b> |
| <b>Bayesian Lasso</b>       | Youden Index     | 0.062            | 0.810    | 0.719       | 0.817       | 0.220     | 10,729        |
| <b>Bayesian Lasso</b>       | Cost-Sensitive   | 0.072            | 0.852    | 0.666       | 0.865       | 0.262     | <b>10,454</b> |
| <b>Bayesian Two-Stage</b>   | Youden Index     | 0.067            | 0.841    | 0.678       | 0.853       | 0.248     | 10,553        |
| <b>Bayesian Two-Stage</b>   | Cost-Sensitive   | 0.070            | 0.851    | 0.665       | 0.864       | 0.260     | <b>10,502</b> |

Table 3: Threshold-Dependent Performance on Hold-Out Test Set ( $N = 30,000$ ).

## 6 Comparative Discussion

1. All models achieve virtually identical ranking performance ( $\text{AUROC} \approx 83.9\%$ ). Aggregating 15 predictors into 3 macro domain scores in the Two-Stage framework results in no loss of predictive power.
2. The Bayesian Lasso yields negligible gains over classical MLE. Given the large sample size ( $N = 120,000$ ) relative to the feature count ( $K = 15$ ), baseline MLE does not overfit, leaving regularization priors largely inactive.
3. Adjusting the decision boundary from the standard 0.50 to  $c^* \approx 0.070 - 0.072$  accounts for default imbalance and asymmetric losses. This shift reduces total financial costs by over 250–300 units compared to Youden-optimized thresholds.
4. While discriminative performance is equivalent across frameworks, the Two-Stage Bayesian architecture offers key operational advantages. It replaces a complex 15-variable vector with three domain scores (*Past Delinquency*, *Financial Capacity*, and *Demographics*), facilitating regulatory compliance, committee reporting, and model auditability.

## 7 Conclusions

This project demonstrates that Bayesian credit scoring models provide well-calibrated risk estimates while offering clear structural advantages over traditional logistic baselines. The Novel Two-Stage Bayesian framework achieves predictive parity with full-space models while providing transparent, domain-level risk breakdowns.